Msc. Data Science Applied Machine Learning Capstone Project

Mortality Prediction In Patients With Acute Kidney Injury (AKI)

Link to colab:
https://colab.research.google.com/drive/1pyeVFrx4ISuciS4riEwRkuN1Mta82LTW

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Introduction and Objectives

Introduction: Acute Kidney Injury (AKI) is a critical condition commonly observed in ICU patients, with significant implications for in- hospital mortality. This project aims to leverage clinical data to predict mortality outcomes, enabling improved decision-making and resource allocation in intensive care units.

Objectives

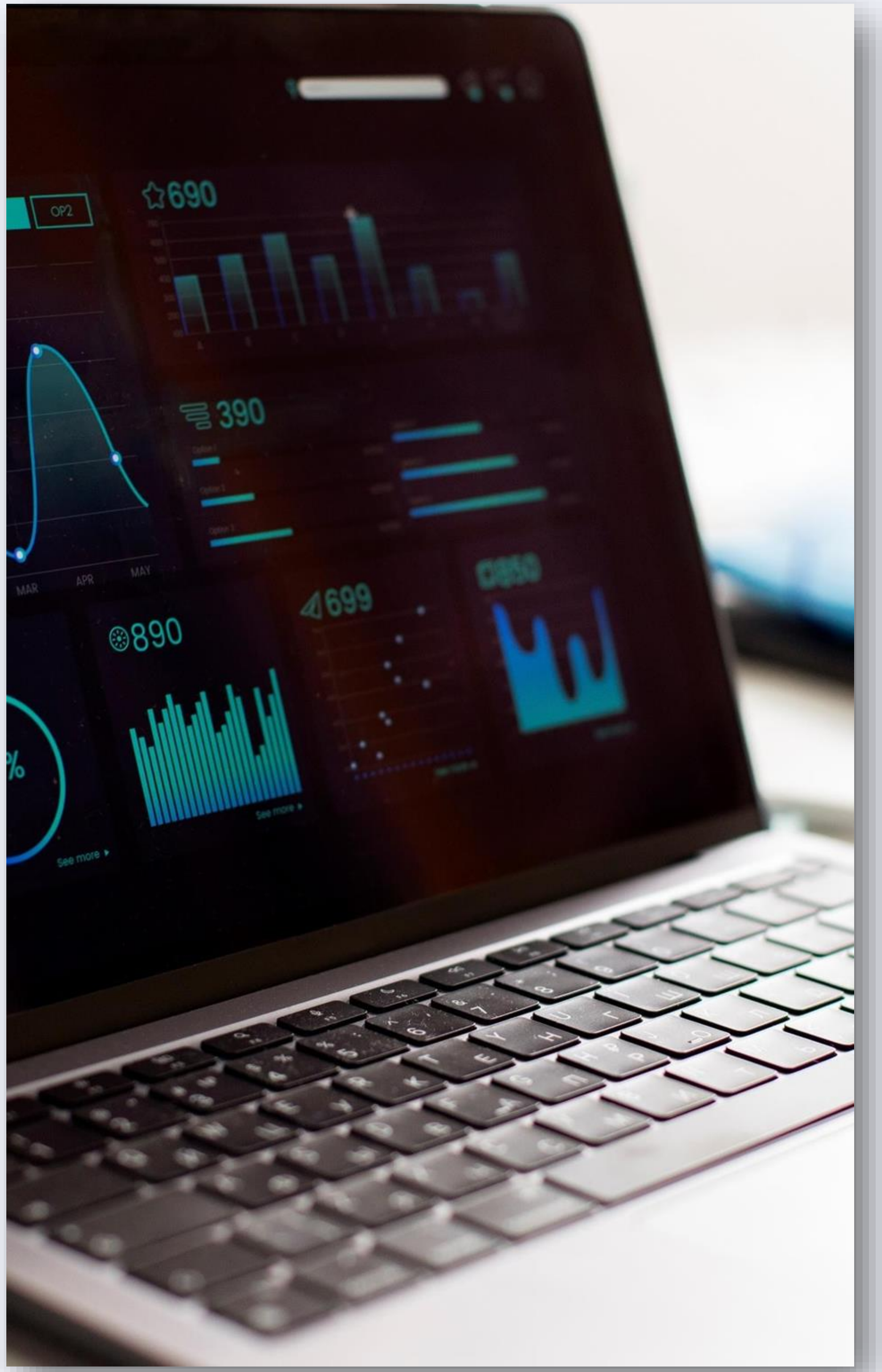
- **To develop a robust predictive model to accurately estimate in-hospital mortality for ICU patients diagnosed with Acute Kidney Injury (AKI)**
- **To identify and prioritize key clinical variables that significantly influence mortality outcomes, supporting early intervention strategies.**
- **To enhance understanding of AKI trajectories, to aid clinicians in resource allocation and personalized patient care.**
- **To provide a foundation for integrating predictive analytics into ICU decision-making processes to improve patient outcomes.**

About the dataset

The dataset is sourced from the MIMIC-III database, a publicly available repository of de-identified health data from ICU patients. It includes detailed clinical variables capturing the progression of AKI in the ICU, such as vital signs, laboratory results, medication use, and comorbidities.

The **target variable, in-hospital mortality (IMH), is binary (1 = mortality, 0 = survival),** enabling the development of classification models to predict outcomes based on patient trajectories.

Key variables



Description of key variables in the data

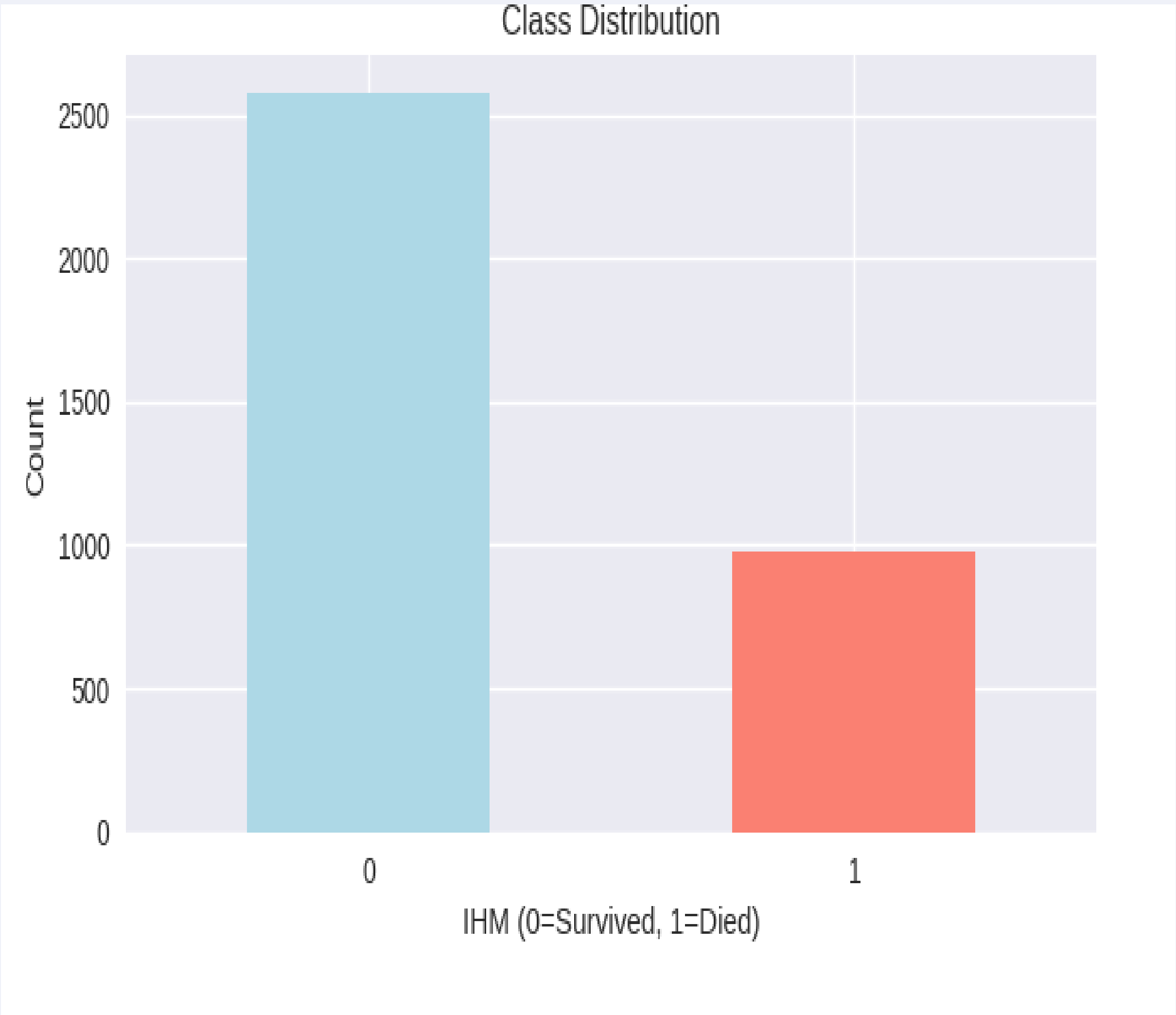
- **Age & Sex:** Demographic information providing essential context for understanding patient profiles and their potential impact on mortality risk.
- **Days in ICU:** The duration of a patient's stay ing the ICU, reflecting the severity and progression of their condition.
- **BP & HR:** Blood pressure and heart rate, vital indicators of cardiovascular stability and overall physiological status.
- **Temp & WBC:** Body temperature and white blood cell count, offering insights into immune activity and the presence of systemic infections.
- **PaO2 & FiO2:** Partial pressure of oxygen in arterial blood and fraction of inspired oxygen, critical metrics for assessing respiratory function and oxygenation efficiency.
- **Pot, Bic, Na:** Levels of potassium, bicarbonate, and sodium, essential electrolytes for maintaining cellular and organ function.
- **BUN & Bilirubin:** Blood urea nitrogen and bilirubin levels, serving as markers of kidney and liver function, respectively.
- **GCS:** Glasgow Coma Scale, a standardized tool for evaluating the level of consciousness and neurological status.
- **IMH:** In-hospital mortality, the target variable for this analysis; indicating survival (0) or mortality (1) outcomes during the ICU stay.

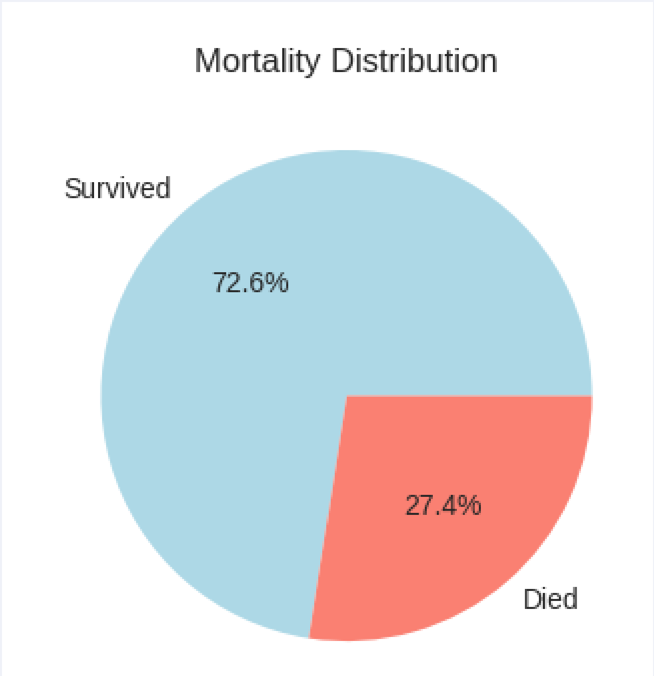
Exploratory data analysis

Aimed to understand the dataset's structure, identify patterns, and prepare for modeling. Key clinical and demographic variables were analyzed for distributions to detect trends, outliers, and skewness. A correlation matrix highlighted relationships and multicollinearity, informing feature selection and engineering. This approach ensured a focus on relevant predictors, laying a solid foundation for preprocessing and predictive modeling

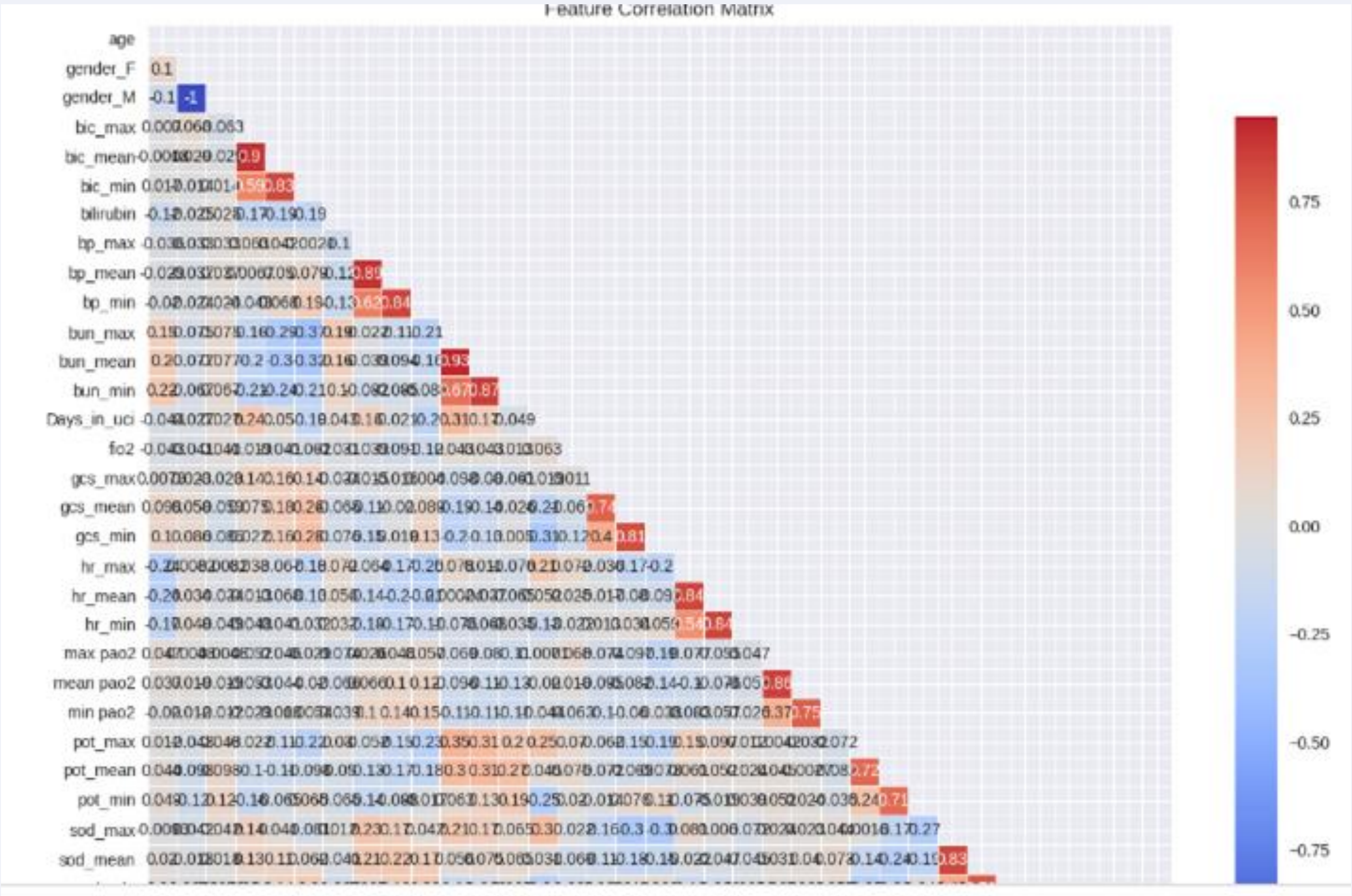
Target Variable Analysis

***Aim:** Visualize the distributions of Mortality*



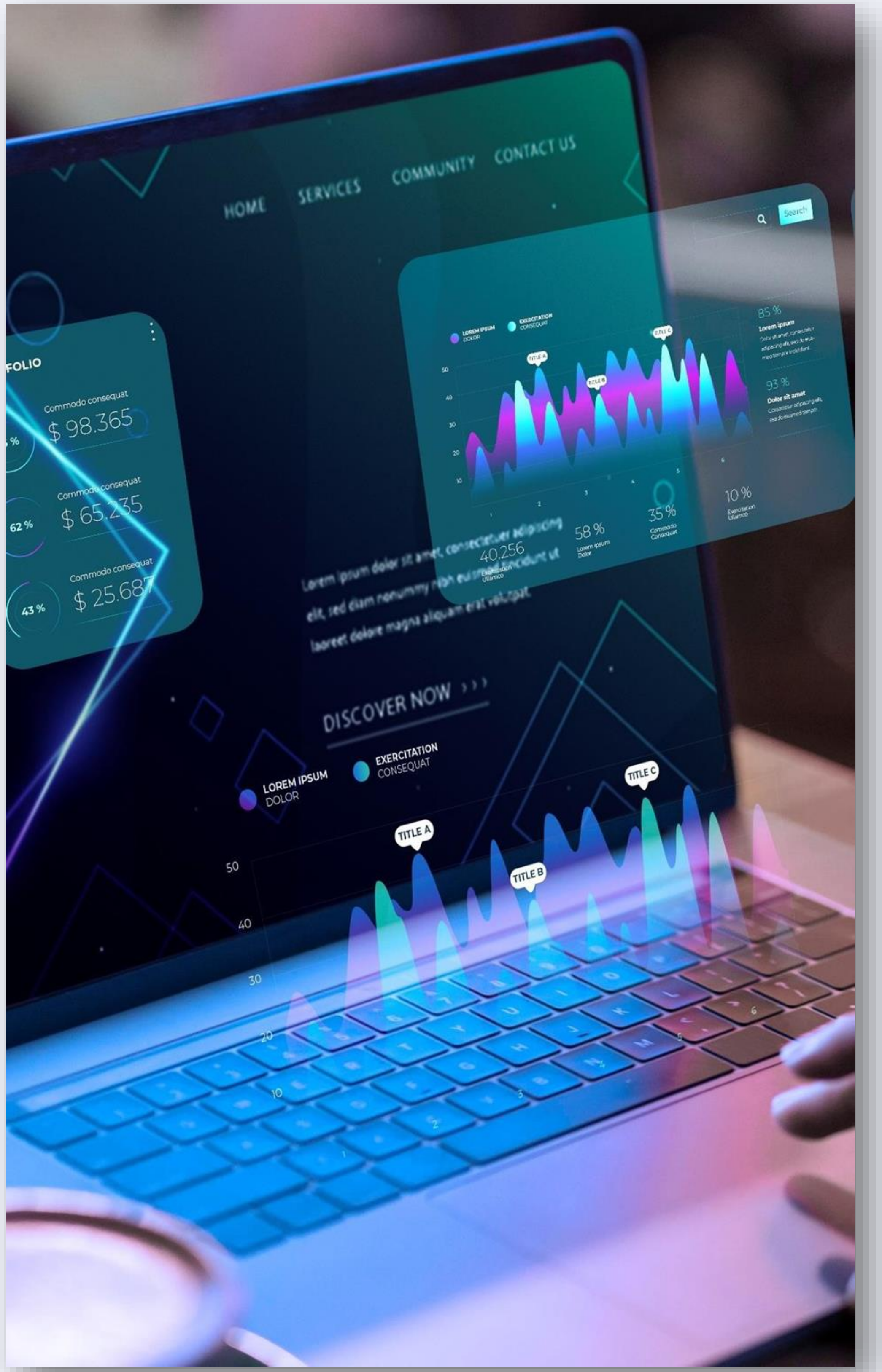


Correlation matrix



Observation: As expected, higher values for Days_in_uci, bun_max, and bun_mean are positively correlated with mortality. Conversely, higher gcs_min and gcs_mean (better consciousness) are negatively correlated with mortality. This aligns with clinical intuition

Data Preprocessing



Preprocessing steps taken

- **Handling Missing Values:** Median imputation was applied to ensure robustness and avoid data loss.
- **Handling Outliers:** May represent critical medical conditions
- **Feature Scaling:** Important for algorithms sensitive to scale
- **Handling Class Imbalance:** Critical in medical prediction tasks

Feature Engineering- key highlight

- The PaO_2/FiO_2 Ratio was introduced to enhance the model's predictive power by providing a nuanced measure of oxygenation efficiency in the lungs. As a key marker of respiratory function, it combines PaO_2 and FiO_2 to assess conditions like Acute Respiratory Distress Syndrome (ARDS) and improve insights into respiratory health, ultimately aiding in more accurate prediction of patient outcomes. It reflects the efficiency of oxygen transfer in the lungs.
- By integrating this ratio, the model gains a feature that directly correlates with key clinical outcomes, enabling it to detect subtle patterns and variations in oxygenation that may signal worsening or improving conditions.

Multiple Algorithm Approach

We trained multiple algorithms to find the best performer:

1. **Logistic Regression**: Interpretable baseline
2. **Random Forest:** Ensemble method, handles non-linearity
3. **Gradient Boosting:** Advanced ensemble technique
4. **Support Vector Machine:** Effective for high-dimensional data
5. **K-Nearest Neighbors:** Instance-based learning
6. **Naive Bayes:** Probabilistic classifier
7. **Deep Neural Network:** Captures complex patterns

MODELS AND THEIR PERFORMANCE

```
=== TRAINING TRADITIONAL ML MODELS ===

Training Logistic Regression...
  Original - F1: 0.651, AUC: 0.873
  SMOTE - F1: 0.680, AUC: 0.876

Training Random Forest...
  Original - F1: 0.643, AUC: 0.874
  SMOTE - F1: 0.685, AUC: 0.873

Training Gradient Boosting...
  Original - F1: 0.655, AUC: 0.885
  SMOTE - F1: 0.694, AUC: 0.878

Training SVM...
  Original - F1: 0.661, AUC: 0.874
  SMOTE - F1: 0.689, AUC: 0.874

Training K-Nearest Neighbors...
  Original - F1: 0.588, AUC: 0.823
  SMOTE - F1: 0.597, AUC: 0.806

Training Naive Bayes...
  Original - F1: 0.579, AUC: 0.824
  SMOTE - F1: 0.618, AUC: 0.824

Training Decision Tree...
  Original - F1: 0.576, AUC: 0.708
  SMOTE - F1: 0.582, AUC: 0.715
```

- **Gradient Boosting** achieved the **highest AUC (0.885)** on the original data and remained strong after SMOTE.
- **SMOTE generally improved the F1 score** across models, confirming that balancing the dataset helped minority class recognition.
- **Decision Tree** and **KNN** performed weakest in AUC, indicating limited discriminative power.

Deep Neural Network (DNN)

```
=== BUILDING DEEP NEURAL NETWORK ===
Training DNN on original data...
Training DNN on SMOTE data...
23/23 [==============================] - 0s 2ms/step
23/23 [==============================] - 0s 2ms/step
23/23 [==============================] - 0s 2ms/step
23/23 [==============================] - 0s 2ms/step
DNN Original - F1: 0.705, AUC: 0.887
DNN SMOTE - F1: 0.709, AUC: 0.871
```

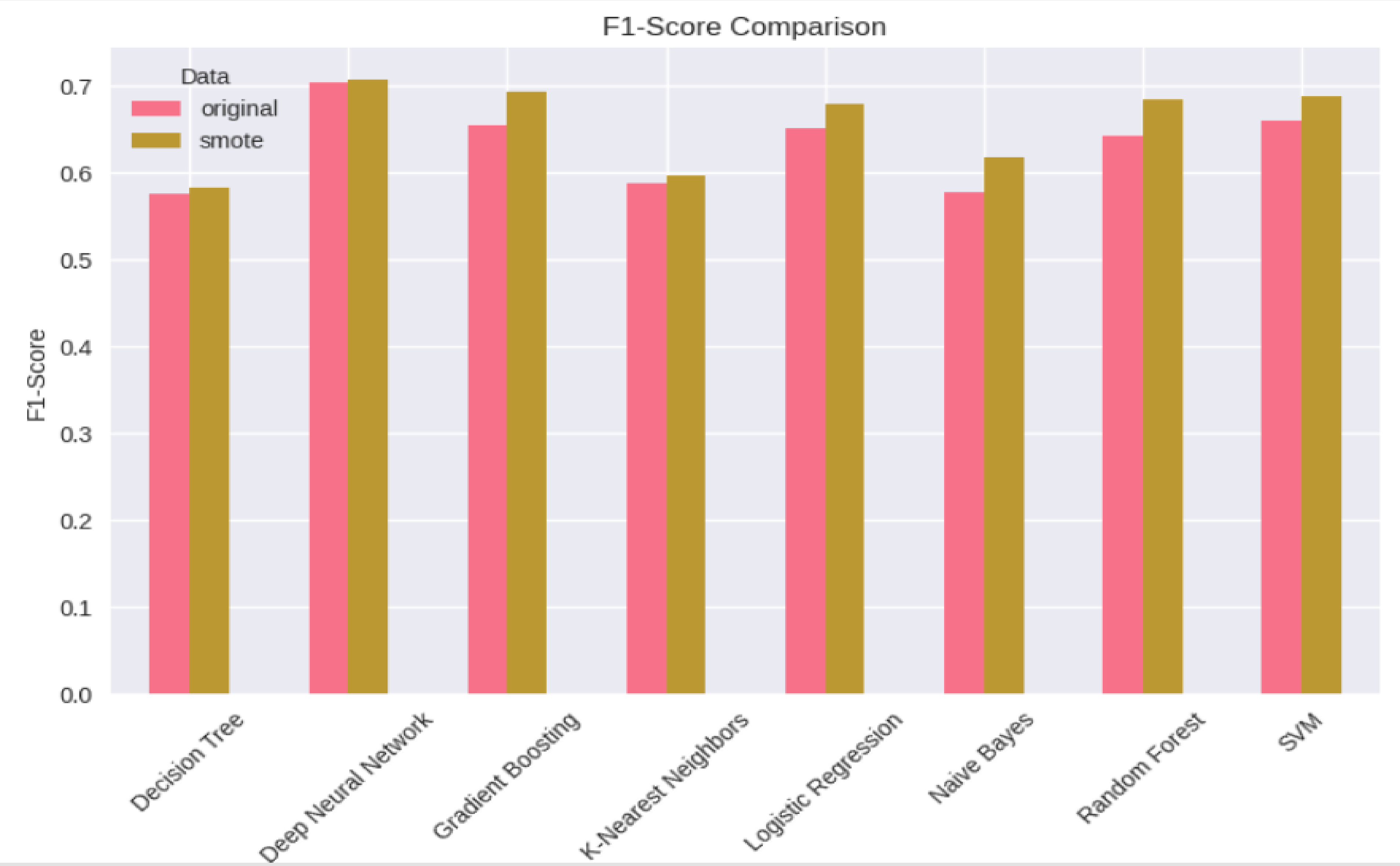
The **Deep Neural Network (DNN)** shows an **F1 score of around 0.70** in both cases, indicating a strong balance between precision and recall.

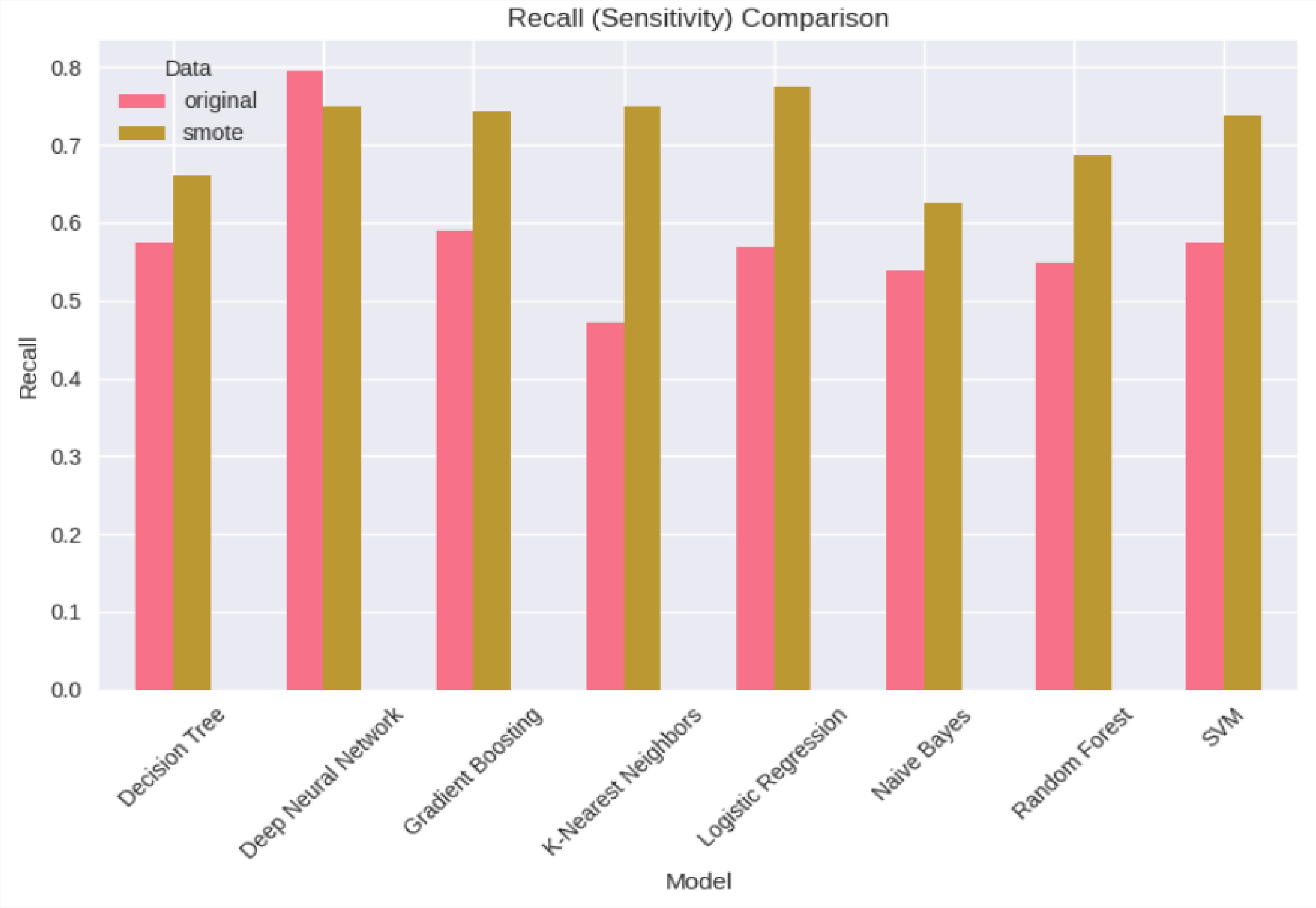
The **AUC of 0.887** on the original data suggests that the model has a high ability to discriminate between the positive and negative classes.

After applying **SMOTE**, the **F1 score improves slightly (0.705 → 0.709)**, showing that class balancing helped the model identify the minority class better.

However, **AUC decreases slightly (0.887 → 0.871)** — a common occurrence because SMOTE can sometimes introduce synthetic noise, slightly reducing the model's overall discriminatory power.

MODEL COMPARISON





SMOTE (Synthetic Minority Oversampling Technique) effectively improved class balance, allowing models to better capture minority class patterns.

Deep learning and ensemble models (DNN, Gradient Boosting, Random Forest) show the **best adaptability** to balanced data.

Simpler models benefit but remain less powerful in complex feature spaces.

SMOTE (Synthetic Minority Over-sampling Technique) enhanced class balance, leading to **better F1 performance across all models**.

Deep Neural Network, followed by **Gradient Boosting** and **Random Forest**, achieved the **best trade-off between precision and recall**, making them ideal candidates for production deployment in this classification task.

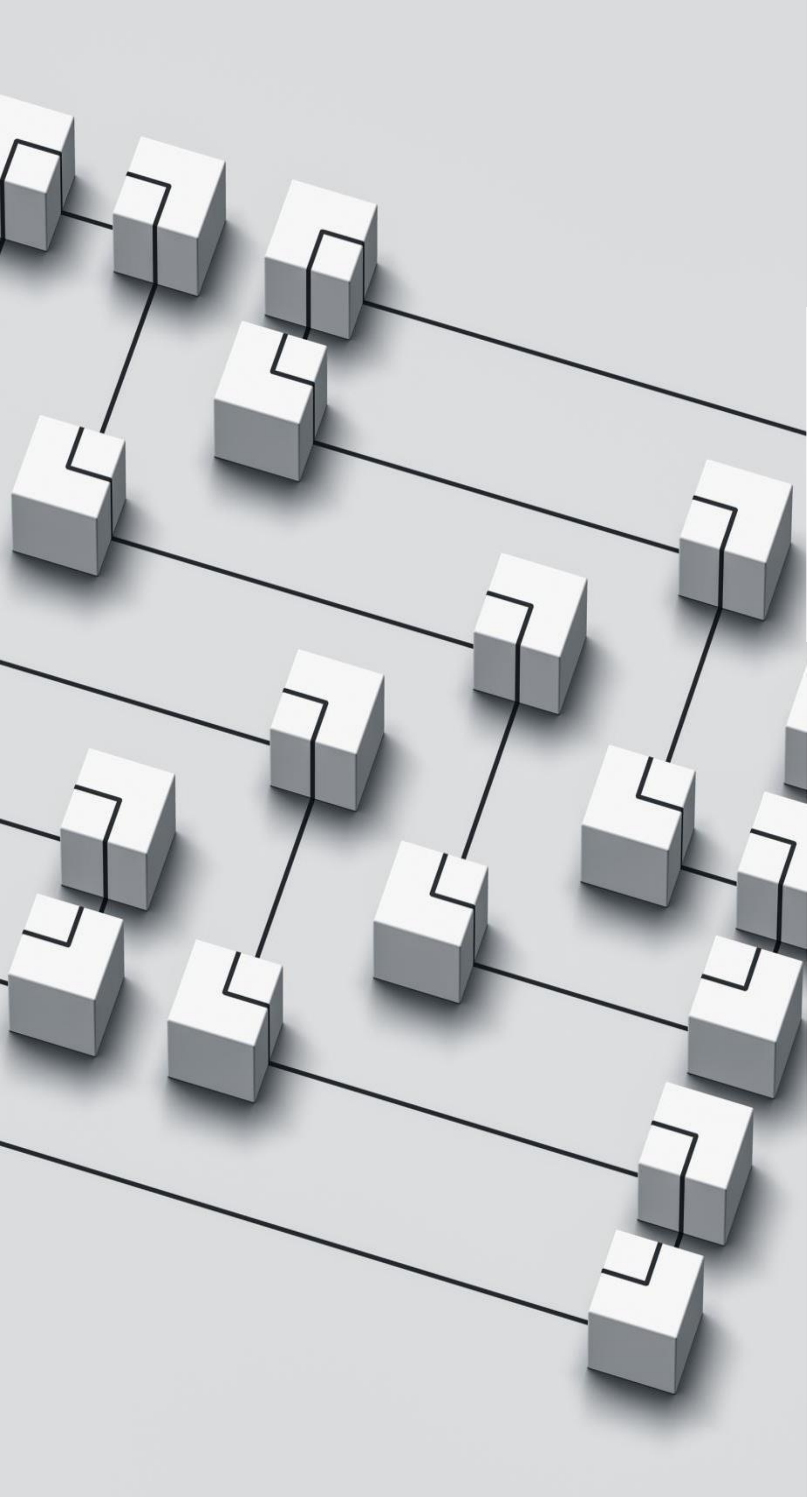
The chart clearly demonstrates that **addressing data imbalance is critical** to improving model reliability and fairness.

MODEL COMPARISON

=== MODEL PERFORMANCE COMPARISON ===

	Model	Data	Accuracy	Precision	Recall	F1-Score	AUC-ROC
0	Logistic Regression	original	0.832	0.760	0.569	0.651	0.873
1	Logistic Regression	smote	0.800	0.606	0.774	0.680	0.876
2	Random Forest	original	0.832	0.775	0.549	0.643	0.874
3	Random Forest	smote	0.827	0.684	0.687	0.685	0.873
4	Gradient Boosting	original	0.830	0.737	0.590	0.655	0.885
5	Gradient Boosting	smote	0.820	0.650	0.744	0.694	0.878
6	SVM	original	0.838	0.778	0.574	0.661	0.874
7	SVM	smote	0.817	0.646	0.738	0.689	0.874
8	K-Nearest Neighbors	original	0.818	0.780	0.472	0.588	0.823
9	K-Nearest Neighbors	smote	0.723	0.497	0.749	0.597	0.806
10	Naive Bayes	original	0.785	0.625	0.538	0.579	0.824
11	Naive Bayes	smote	0.787	0.610	0.626	0.618	0.824
12	Decision Tree	original	0.768	0.577	0.574	0.576	0.708
13	Decision Tree	smote	0.739	0.520	0.662	0.582	0.715
14	Deep Neural Network	original	0.817	0.633	0.795	0.705	0.887
15	Deep Neural Network	smote	0.831	0.673	0.749	0.709	0.871

- **Deep Neural Network** achieved the **best overall balance** — highest F1 (0.709) and AUC (0.887 before SMOTE), showing strong predictive power and generalization.
- **Gradient Boosting and Random Forest** followed closely, performing robustly on the datasets.
- **SMOTE** effectively improved recall and F1, validating the importance of class balance in imbalanced classification tasks.



BEST PERFORMING MODELS

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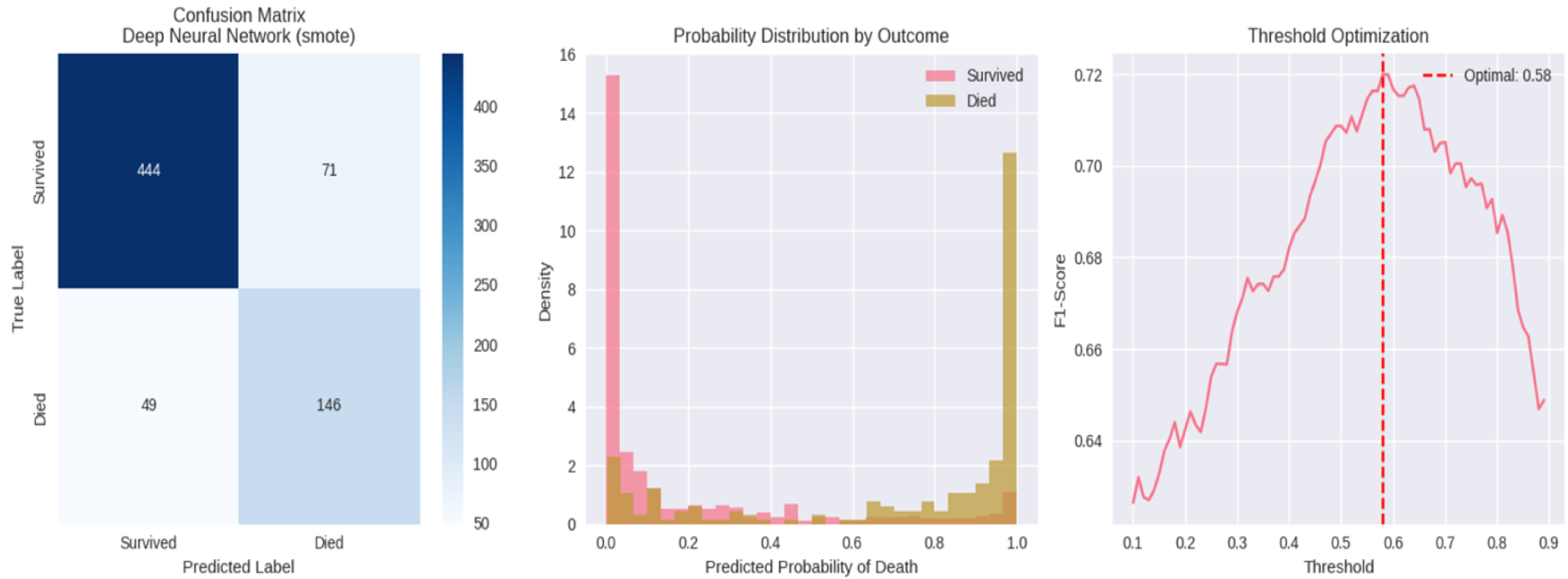
=== BEST PERFORMING MODELS ===
Best F1-Score: Deep Neural Network (smote) - 0.709
Best AUC-ROC: Deep Neural Network (original) - 0.887
Best Recall: Deep Neural Network (original) - 0.795
```

The **Deep Neural Network** clearly **outperformed all traditional machine learning models** across major metrics.

SMOTE improved F1 slightly, helping balance predictions between precision and recall.

However, the **original dataset version of the DNN maintained the best AUC and recall**, showing it generalized well without needing synthetic samples.

FINAL MODELS AND THEIR PERFORMANCE



The model correctly classified **444 survivors** and **146 deaths**.

- It made **71 false positives** (predicted "died" but actually survived) and **49 false negatives** (predicted "survived" but actually died).
- These are **good results**, showing a strong balance between sensitivity and specificity.

Model Strength: High true positive and true negative rates.
Model Weakness: Some confusion between the two classes — mainly in predicting "died" cases.

The two distributions are **well-separated**, meaning the model effectively differentiates between survivors and deaths.

Only a small overlap occurs in the middle (around 0.4–0.6), where predictions are uncertain.

The **default threshold (0.5)** is not necessarily optimal; at **0.58**, the model achieves the **highest F1 (~0.71)**.

This means setting the cutoff slightly higher improves the trade-off between precision and recall, reducing false positives while maintaining good sensitivity.

Clinical Insights and Feature Analysis

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=== MODEL INTERPRETATION ===
The Deep Neural Network model shows:
• Overall accuracy of 83.1%
• Sensitivity of 74.9% (correctly identifies 74.9% of patients who will die)
• Specificity of 86.2% (correctly identifies 86.2% of patients who will survive)
• F1-score of 0.709 (balanced precision and recall)
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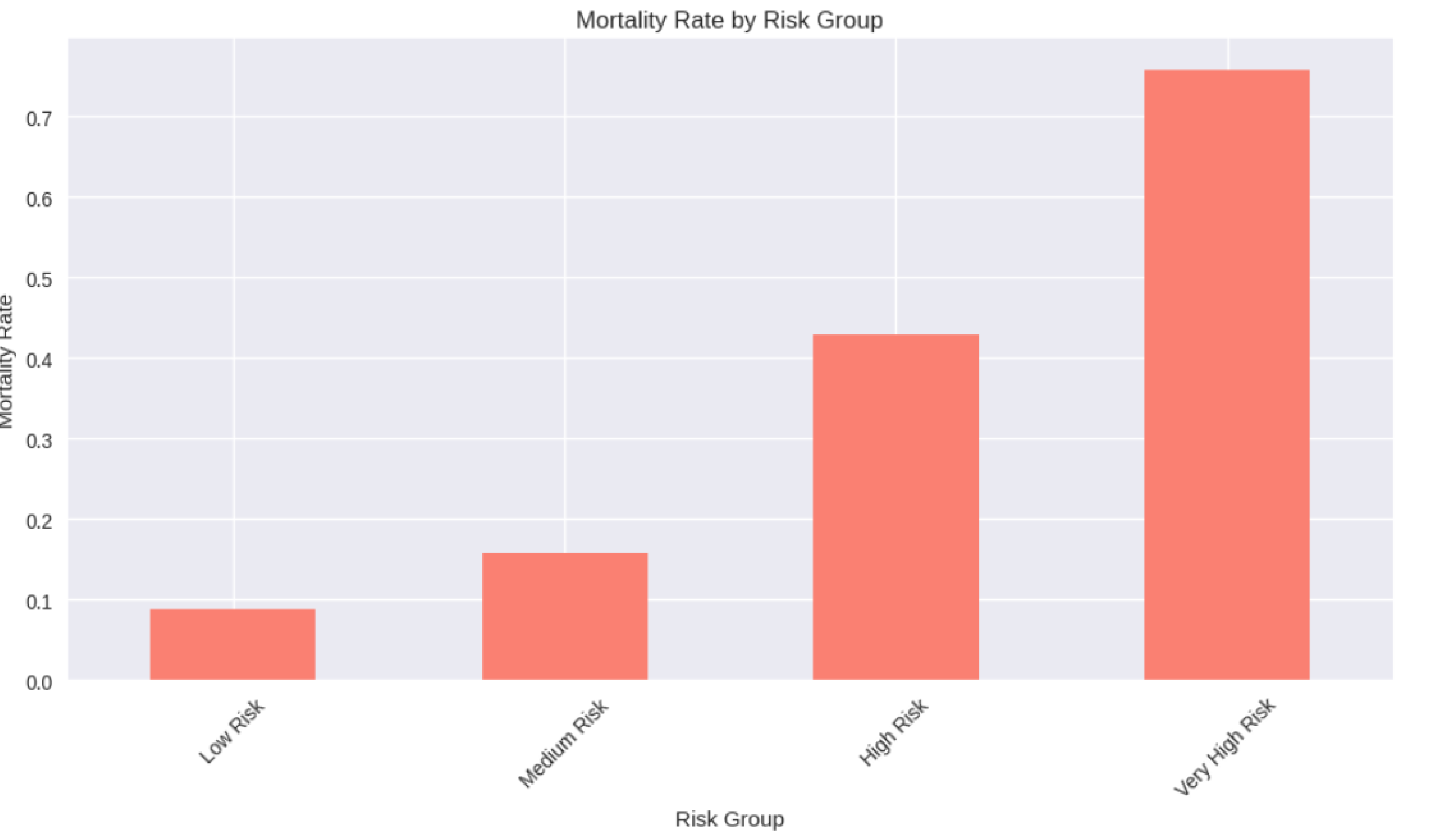
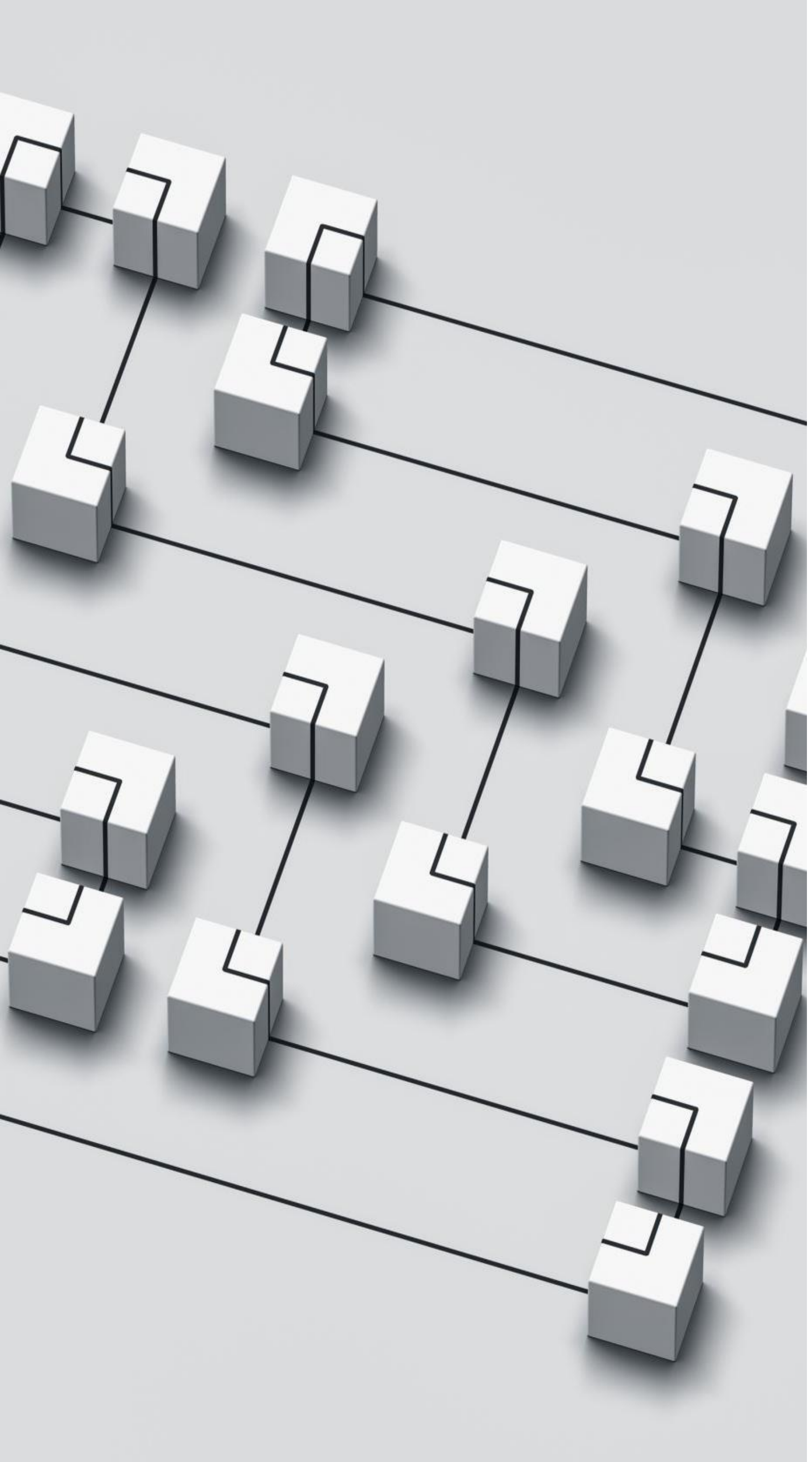


The model demonstrates **excellent discriminative ability**: it is sensitive enough to detect most at-risk patients while maintaining high specificity for survivors.

The **F1-score of 0.709** shows a good balance between false positives (predicting death when alive) and false negatives (predicting survival when death occurs).

These metrics, combined with the **risk group chart**, confirm that the DNN not only predicts outcomes correctly but also ranks patients' mortality risk realistically.



SAVING THE BEST MODEL

The **Deep Neural Network (DNN)** achieved the **highest performance** (F1 = 0.709, AUC = 0.887).

This model was selected as the **final production model** due to its superior balance between precision and recall.Deep Neural Network (DNN):

- Method: `joblib.dump(best_dnn_model, "best_dnn_model.pkl")`.
- File Format: Pickle (`.pkl`).
- Library: `joblib` is lightweight and efficient for serializing Python objects, especially tree- based models.
- Reloading: Use `joblib.load("best_dnn_model.pkl")` to restore the model.

Advantages of Model Serialization:

- Portability: Enables moving models across environments (e.g., development to production).
- Efficiency: Saves time by avoiding retraining.
- Scalability: Facilitates deployment in real-world applications.

Challenges

- **Imbalanced Dataset:**
  The “died” class had far fewer observations compared to the “survived” class, leading models to favor the majority class.
- **Data Quality Issues:**
  Missing, noisy, or inconsistent patient records (e.g., missing lab values, categorical inconsistencies) can reduce predictive accuracy.
- **Feature Redundancy or Irrelevance:**
  Some features might not contribute meaningfully to predicting mortality, creating noise in training.
- **Small Dataset Size:**
  Deep Neural Networks typically need large datasets to generalize well; limited data can lead to overfitting.

SOLUTIONS ADAPTED

To overcome the challenges, several strategies were implemented:

- **Balancing Techniques:**
 Used **SMOTE (Synthetic Minority Oversampling Technique)** to generate synthetic examples of minority cases and improve minority-class recall.
- **Data Cleaning & Imputation:**
 Applied preprocessing steps to handle missing values, remove duplicates, and standardize features.
- **Feature Selection/Engineering:**
 Used correlation checks, domain knowledge, and feature importance analysis to select the most relevant variables.
- **Regularization & Dropout:**
 Added dropout layers and L2 regularization in the DNN to mitigate overfitting due to limited samples.

Conclusion

- Developed a robust **mortality prediction model** using traditional ML and **Deep Neural Networks (DNN)**.
- The **DNN (with SMOTE)** achieved the **best overall performance** —
  Accuracy: 83.1% | F1: 0.709 | AUC: 0.887 | Sensitivity: 74.9% | Specificity: 86.2%.
- **SMOTE** improved class balance and recall for minority (death) cases.
- **Risk stratification** clearly separated patients into Low, Medium, High, and Very-High mortality groups.
- Visual tools (confusion matrix, probability plots, threshold tuning) enhanced **model interpretability** and **clinical relevance**.

Future Works & Recommendations

- **External Validation:**
  Test the model on independent hospital datasets to confirm generalizability.

- **Model Deployment:**
  Build an **interactive dashboard or API** for real-time risk prediction.

- **Continuous Learning:**
  Implement model retraining pipelines to update predictions as new data arrive.

- **Ethical Oversight:**
  Monitor for bias, ensure fairness across demographic groups, and maintain data privacy compliance.

MEMBER CONTRIBUTIONS

- **Alex Gikaru** – Introduction and Objectives (Overview of the project and dataset), EDA, Data preprocessing
- **Kevin Obote** – Model development & tuning
- **Kennedy Mwenda** – Evaluation & visualization
- **Adeline Makokha** – Challenges and Solutions, Conclusion